Approximate Bayesian Inference

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Outline

Approximate Bayesian Inference

Monte-Carlo Methods

Packages

Approximate Bayesian inference

The General problem

- Observations D.
- Nuisance variables z.
- Unknown parameter θ.
- Direct calculation of any of these terms can be infeasible:

$$\phi(\theta \mid D) = \frac{P_\theta(D)\phi(\theta)}{\sum_{\theta'} P_{\theta'}(D)\phi(\theta')}, \qquad P_\theta(D) = \sum_z P_\theta(D, z).$$

Common methods

- Monte Carlo
- Variational Bayes
- Approximate Bayesian Computation (ABC)
- Stochastic Variational Inference

The Monte-Carlo sampler

To approximate an expectation, we can approximate the integral

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega) dP(\omega)$$

by sampling ω from P and taking the sample mean:

$$\mathbb{E}_P(f) \approx \frac{1}{N} \sum_{n=1}^{N} f(\omega^{(n)}),$$

which as we already know, is an unbiased estimator.
We can also approximate the probabilities of events $A \subset \Omega$:

$$P(A) = \int_A dP(\omega) = \int_\Omega \mathbb{I}\{\omega \in A\} dP(\omega) \tag{1}$$

$$= \mathbb{E}_P[\mathbb{I}\{\omega \in A\}] \approx \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}\left\{\omega^{(n)} \in A\right\}. \tag{2}$$

But what if we can not sample from P?

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But what if we can not sample from P?

Notes.

The sample mean

- If $x_n = f(\omega^{(n)})$ this is the sample mean $\frac{1}{N} \sum_{n=1}^{N} x_n$.
- Example: ω is a randomly chosen person and f is the height measurement.

Sampling

- In the data analysis setting, the samples come from nature's unknown distribution P. Here, however we are using sampling to perform approximate computation. Nevertheless, the mathetmatical tools are the same.
- Because we define the sampling, we can decide what distribution to sample from. But how do we sample something random in the first place?
- If we have a fair coin, we can get n random bits, to obtain a number in $\{0, 1, \ldots, 2^n - 1\}$. Dividing by 2^n, we can approxiamte the uniform distribution in $[0, 1]$.
- In practice, we only have a few random bits, which we combine with deterministic algorithms (Pseudo-Random Number Generators) to get the illusion of randomness.

Importance sampling

If we cannot sample from P, but can sample from some other distribution Q, we can use this trick[1]. Let's write it first out for finite Ω, for simplicity

$$\mathbb{E}_P(f) \triangleq \sum_{\omega \in \Omega} f(\omega) \frac{P(\omega)}{Q(\omega)} Q(\omega) = \mathbb{E}_Q(fP/Q)$$

Now this is the expectation of $f(\omega)P(\omega)/Q(\omega)$ with respect to Q. So, if we can sample from Q instead, we can still approximate the integral for the general case.

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega) \frac{dP(\omega)}{dQ(\omega)} dQ(\omega) \approx \frac{1}{N} \sum_{n=1}^{N} f(\omega^{(n)}) \frac{dP(\omega^{(n)})}{dQ(\omega^{(n)})}, \qquad \omega^{(n)} \sim Q.$$

It is easy to show that this estimator is also unbiased. However, as the probability ratio can be very large, it has a very high variance.
So, how can we sample from P for arbitrary distributions?

[1] Statisticians hate it!

Basic sampling theory

Inversion sampler

$F(u) \triangleq \mathbb{P}(x \leq u) = P(\{\omega : x(\omega) \leq u\})$ is the CDF of x.

- Sample u uniformly in $[0, 1]$
- Set $x = F^{-1}(u)$, i.e. $v : \mathbb{P}(x \leq v) = u$

Rejection Sampler

- Input: $\epsilon > 0$, distribution Q with $\epsilon P \leq Q$.
- Repeat:
- $\hat{x} \sim Q$.
- $u \sim \mathrm{Unif}[0, 1]$
- Until $u \leq \epsilon P(\hat{x})/Q(\hat{x})$.
- Return $\hat{x}$

Notes

- Useful for sampling from a known distribution P.
- A variant of rejection is used for ABC (see later).

Monte-Carlo sampling

Approximating a density is tricky, so we will instead approximate the posterior probability that the parameter is in $B \subset \Theta$:

$$\phi(B \mid D) = \int_B d\phi(\theta|D) = \frac{\int_B P_\theta(D) d\phi(\theta)}{\int_\Theta P_{\theta'}(D) d\phi(\theta')} = \frac{\int_B P_\theta(D) d\phi(\theta)}{\int_\Theta P_{\theta'}(D) d\phi(\theta')}$$

We can approximate the integrals by sampling from the prior ϕ:

$$\int_B P_\theta(D) d\phi(\theta) \approx \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}\left\{\theta^{(n)} \in B\right\} P_{\theta^{(n)}}(D), \qquad \theta^{(n)} \sim \phi.$$

- Sampling from the prior is inefficient: almost all the samples land where the likelihood is near 0.
- The estimator has high bias and variance, because the denominator is also an estimate.
- So, we can use Markov Chain Monte Carlo. This lets us sample a sequence $\theta^{(n)}$ which converges asymptotically[2] to $\phi(\theta^{(n)}|D)$.

[2] You don't have to worry about asymptotics in this course.

Markov Chain Monte Carlo

Recall the chains from the graphical model lecture. These model a sequence of random variables so that $\mathbb{P}(x_{t+1} \mid x_t, x_{t-1}, \ldots, x_1) = \mathbb{P}(x_{t+1} \mid x_t)$.
Now we are going to generate samples from a Markov chain on the fly. However, we can design the probabilities $\mathbb{P}(x_{t+1} \mid x_t)$ so that the distribution $\mathbb{P}(x_t)$ (for large t) is close to the distribution we want.

MCMC for posterior sampling

- Here we are interested in approximating the distribution $\mathbb{P}(\theta \mid D)$.
- To do this, we form a Markov chain $\mathbb{P}(\theta^{(n+1)} \mid \theta^{(n)}, D)$

MCMC for other latent variables

- Here we are interested in approximating the distribution $\mathbb{P}(z \mid D)$.
- Form a Markov chain $\mathbb{P}(z^{(n+1)} \mid z^{(n)}, D)$

The Gibbs sampler

This is used when we need to sample over only some variables $z_1, \ldots, z_k$, given some fixed variables x.

General algorithm

- Input: Factors $P(z_k \mid z_1, \ldots z_{k-1}, z_{k+1}, \ldots, z_K, x)$
- For $n \in [N]$:
- For $k \in [K]$: $z_k^{(n)} \sim P(z_k \mid z_1^{(n)}, \ldots z_{k-1}^{(n)}, z_{k+1}^{(n-1)}, \ldots, z_K^{(n-1)}, x)$

Application to posterior sampling with latent variables:

Latent variable z, parameter θ.

- Until convergence:
- $\theta^{(n)} \sim P(\theta \mid z^{(n-1)}, x)$
- $z^{(n)} \sim P(z \mid \theta^{(n)}, x)$

Metropolis-Hastings I

Algorithm (symmetric version)

- Input: Proposal distribution $Q(x|x') = Q(x'|x)$
- At time n:
- $\hat{x} \sim Q(x|x^{(n)})$
- w.p. $P(\hat{x})/P(x^{(n)})$, $x^{(n+1)} = \hat{x}$ else $x^{(n+1)} = x^{(n)}$ (capped at 1)

Application to posterior sampling:

The denominator cancels out, leading to:

$$\frac{\phi(\theta' \mid D)}{\phi(\theta \mid D)} = \frac{P_{\theta'}(D)\phi(\theta')}{P_{\theta}(D)\phi(\theta)}$$

The only question is which proposal to use.

Metropolis-Hastings II

Algorithm

- Input: Proposal distribution $Q(x|x')$, likelihood P (up to constant factor)
- At time n:
- $\hat{x}|x^{(n)} \sim Q(x|x^{(n)})$
- With probability (capped at 1)

$$\frac{P(\hat{x})Q(x^{(n)}|\hat{x})}{P(x^{(n)})Q(\hat{x}|x^{(n)})},$$

  set $x^{(n+1)} = \hat{x}$
- Otherwise $x^{(n+1)} = x^{(n)}$

Application to posterior sampling:

The $\mathbb{P}_\phi(D)$ term cancels out, leading to:

$$\frac{\phi(\theta' \mid D)Q(\theta \mid \theta')}{\phi(\theta \mid D)Q(\theta' \mid \theta)} = \frac{P_{\theta'}(D)\phi(\theta')Q(\theta \mid \theta')}{P_\theta(D)\phi(\theta)Q(\theta' \mid \theta)}$$

M-H Theory

Stationary distribution

The Markov chain defined by the M-H algorithm must have a unique stationary distribution

$$\sigma = \sigma \boldsymbol{P},$$

where $\boldsymbol{P}$ is the transition kernel of the chain with

$$P_{ij} = \mathbb{P}(x^{(n+1)} = j \mid x^{(n)} = i).$$

In addition, $\lim_{n\to\infty} \boldsymbol{P}^k = 1\sigma$.

Sufficient conditions

- If the transition kernel satisfies detailed balance:

$$P(x'|x)\sigma(x) = P(x|x')\sigma(x')$$

then $\sigma(x)$ is a stationary distribution.

- If the Markov chain is ergodic then there is a unique σ.

ABC: Approximate Bayesian Computation

In standard probabilistic learning, we compare the probability $P_\theta(D)$ of the data D under different parameters θ. ABC is used when $P_\theta(D)$ is unavailable. There are two settings:

Simulator: we can sample from $P_\theta(D)$.

- An example is a weather model. We can generate data (predictions) from the simulator for various parameters θ.
- Here θ includes e.g. how much light the clouds disperse, or

heat the ocean can store.

- There is no analytic likelihood defined: we randomise by starting the weather model with different random seeds.

Intractable likelihood: hard to calculate $P_\theta(D)$.

There may be a way to calculate the probability, but it is too costly.

General idea

Replace probabilities with a distance between statistics

ABC Rejection sampling

A metric ρ over datasets

- $\rho(D, D')$ is distance between datasets.
- We can use that to define a rejection sampler

ABC Rejection Sampling

- **Input**: $\epsilon > 0$.
- Sample $\theta' \sim \phi(\theta)$
- Sample $D' \sim P_{\theta'}$.
- If $\rho(D, D') \leq \epsilon$, accept θ'

Theorem

If $\rho(D, D') = \|f(D) - f(D')\|$ and f is a sufficient statistic and $\epsilon = 0$ then ABC Rejection Sampling is exact.

ABC: a toy example

Data

- $D = (x_1, \ldots, x_n)$
- $x_i \mid \theta \sim \text{Normal}(\theta, 1)$
- prior $\theta \sim \text{Normal}(0, 10)$. (we don't know what θ is.)

ABC parameters

- Statistic $f(D)$ = sample mean (sufficient here);

$$\rho(D, D') = |f(D) - f(D')|$$

- Repeat: sample θ' from the prior, simulate D', accept if $\rho(D, D') \leq \epsilon$.

Which method when?

- Expectation under P, and you can sample from P: plain Monte Carlo.
- You can evaluate P but not sample from it, and Q is close to P: importance or rejection sampling.
- Posterior far from the prior, or many latent variables: MCMC (Metropolis-Hastings in general; Gibbs when the conditionals are tractable).
- Likelihood intractable but you can simulate data: ABC.
- Variational inference: not mentioned. Uses analytical approximations of the posterior. Fast but can be wrong.

Purpose

- Samples from the posterior: MC, MC-IS, MCMC, ABC.
- Approximate posterior: VI, SVI.
- Point estimate: EM, stochastic gradient (maximum likelihood or MAP).

Combos

- EM with an approximate posterior = variational EM
- SGA + VI = SVI
- GA + MCMC = HMC; SGA + MCMC = SGLD / SGHMC

Packages

Hand-coding MCMC samplers is tricky nad error-prone. There are many packages to choose from.

Multi-platform

- STAN
- BUGS

Python

- PyMC
- TensorFlow Probability
- PyStan
- Pyro (Torch)[3]

[3] Examples in the lab. We use it because it shares commonalities with standard Torch.